THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 2

Chapter 8: **Heredity** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The new combinations of traits (round green and wrinkled yellow) in F2 show that 1

(a) traits are always linked **(b)** no dominant trait exists
(c) F1 gets only one parent's traits **(d)** traits are inherited independently

2. The total number of chromosomes in a human body cell is 1

(a) 44 **(b)** 48 **(c)** 46 **(d)** 23

3. When F1 plants of the tall x dwarf cross are self-pollinated, the F2 phenotypic ratio is 1

(a) 3 tall : 1 dwarf **(b)** 1 : 1 **(c)** all tall **(d)** 1 tall : 3 dwarf

4. Genes are located on 1

(a) ribosomes **(b)** cell wall
(c) mitochondria only **(d)** chromosomes

5. The number of autosomes in a human body cell is 1

(a) 23 **(b)** 46 **(c)** 44 **(d)** 22

6. The genotypic ratio of the F2 generation in a monohybrid cross is 1

(a) 1 : 1 **(b)** 1 : 2 : 1 **(c)** 3 : 1 **(d)** 9 : 3 : 3 : 1

7. The set of genes of an organism is its 1

(a) phenotype **(b)** gamete **(c)** allele **(d)** genotype

8. The probability that a child will be a boy is 1

(a) 75% **(b)** 50% **(c)** 25% **(d)** 100%

9. A trait which expresses itself in the F1 generation is called 1

(a) mutated **(b)** recessive **(c)** acquired **(d)** dominant

10. A plant that has two different alleles for a trait is called 1

(a) heterozygous **(b)** recessive **(c)** homozygous **(d)** pure line

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. State Mendel's law of independent assortment. 2

12. Why are traits like muscle size not passed on to offspring? 2

13. Define dominant and recessive traits. 2

14. How many chromosomes does a human sperm and egg have each? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. List three differences between acquired and inherited traits. 3

16. What are autosomes and sex chromosomes? How many pairs are there in humans? 3

17. How many different types of gametes can a plant with genotype RrYy produce? Name them. 3

OR

A man and a woman have a baby. Explain how the child receives traits from both parents. 3

18. How does the accumulation of variation take place over generations in asexually reproducing organisms? 3

OR

Why can the wrinkled-green seeds appear only in the F2 and not in the F1? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. What is meant by 5
(a) dominant trait,
(b) recessive trait,
(c) homozygous,
(d) heterozygous? Give an example for each.

OR

How is variation produced in a population that reproduces sexually? Explain its importance. 5

20. Explain how sex is determined in human beings. What are the chances of having a boy or a girl? 5

OR

Why are the chromosomes present in pairs? Explain how the number of chromosomes remains constant from generation to generation. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND